

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

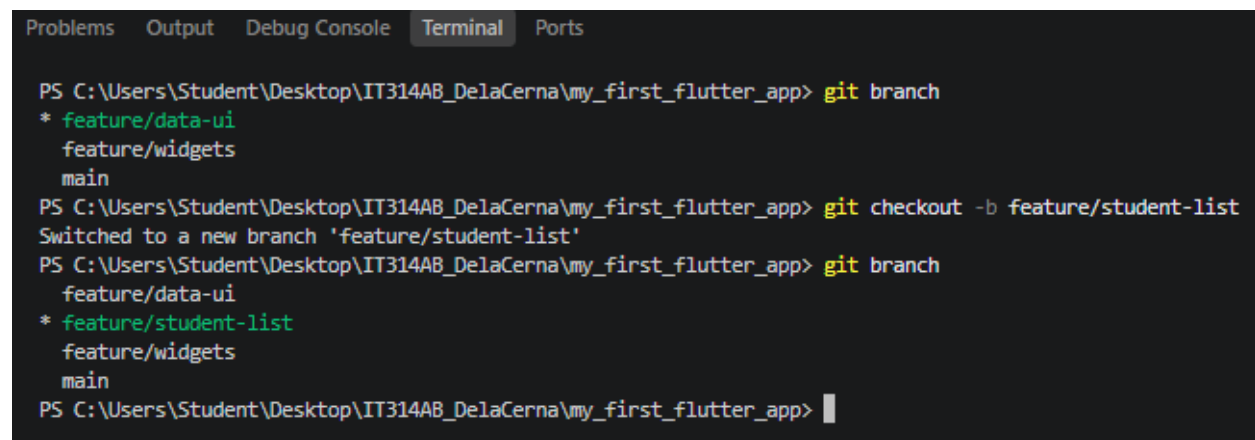
Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git branch
* feature/data-ui
  feature/widgets
  main
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git branch
  feature/data-ui
* feature/student-list
  feature/widgets
  main
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> |
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```
List<Profile> profiles = [
    Profile(
        image: 'assets/DelaCerna.jpg',
        name: 'Leachim Dela Cerna',
        course: 'BSIT',
        yearLevel: 3,
        age: 21,
        hobby: 'Dancing',
    ),

    Profile(
        image: 'assets/barbie.jpg',
        name: 'Julia Kong',
        course: 'BSIT',
        yearLevel: 3,
        age: 20,
        hobby: 'Singing',
    ),

    Profile(
        image: 'assets/barbie1.jpg',
        name: 'Mav Tajanlangit',
        course: 'BSIT',
        yearLevel: 3,
        age: 21,
        hobby: null,
    ),

    Profile(
        image: 'assets/barbie3.jpg',
        name: 'Dom Ocarol',
        course: null,
        yearLevel: 3,
        age: 20,
        hobby: 'Watching Movies',
    ),
]
```

```
Profile(
    image: 'assets/barbie4.jpg',
    name: null,
    course: 'BSIT',
    yearLevel: 3,
    age: 22,
    hobby: 'Gaming',
),
];
```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

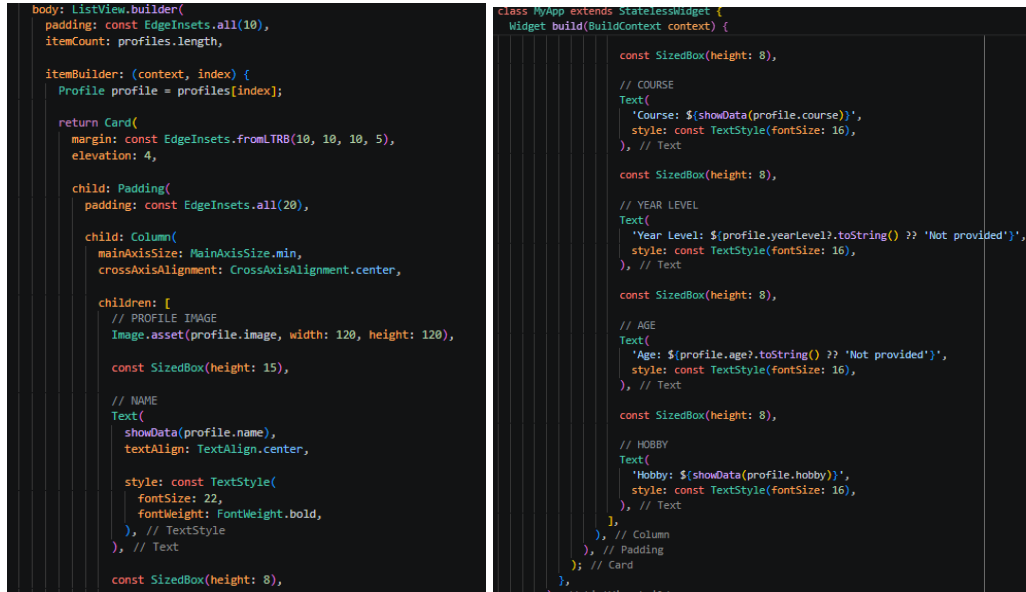
- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)



🚩 Flag 3 — Discover `ListView`

Objective: Make your student directory scrollable using `ListView`.

Research Flutter's `ListView`.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

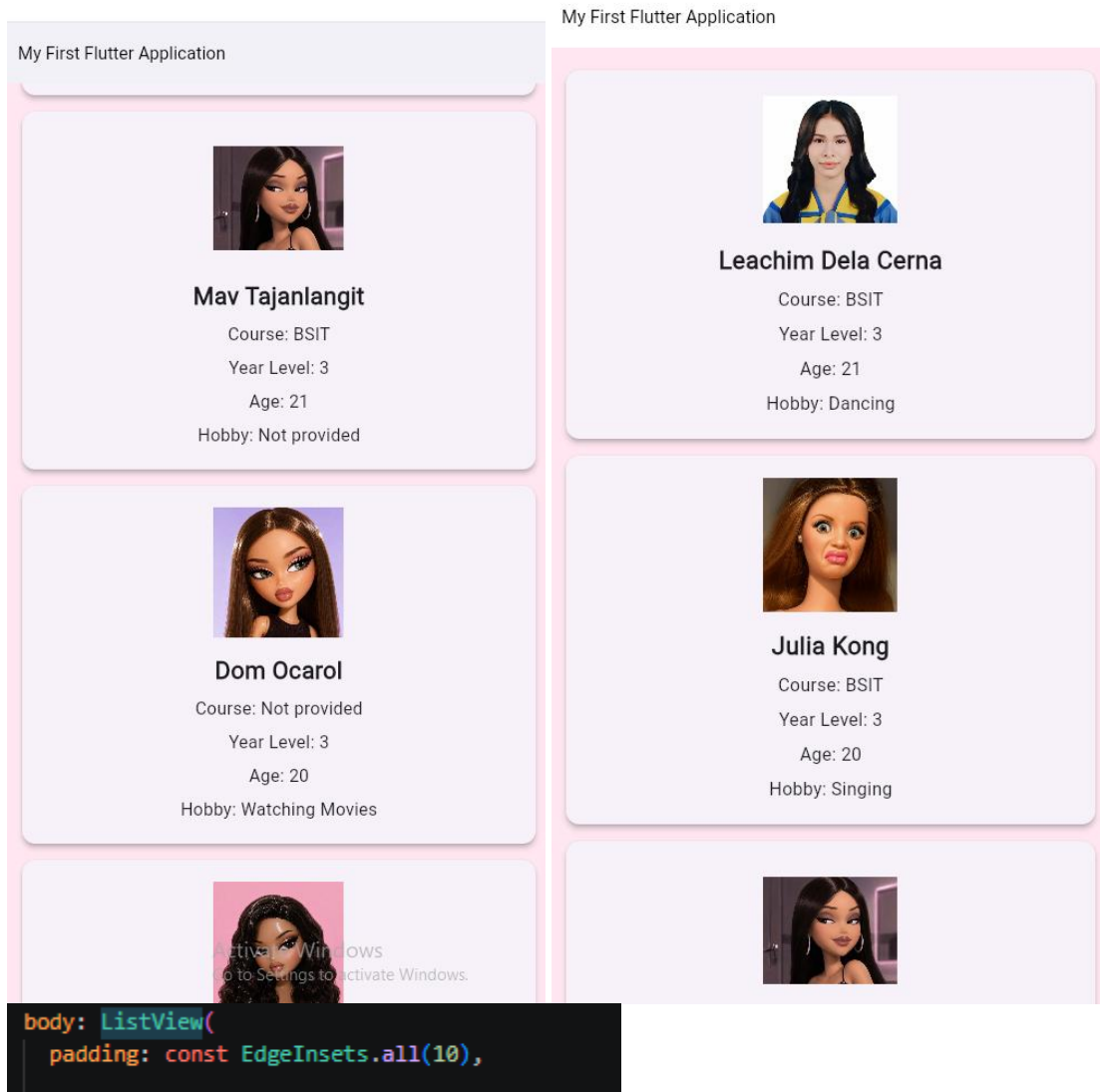
Test

The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- `ListView` in your code
- Top of the application
- Bottom portion showing later students



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

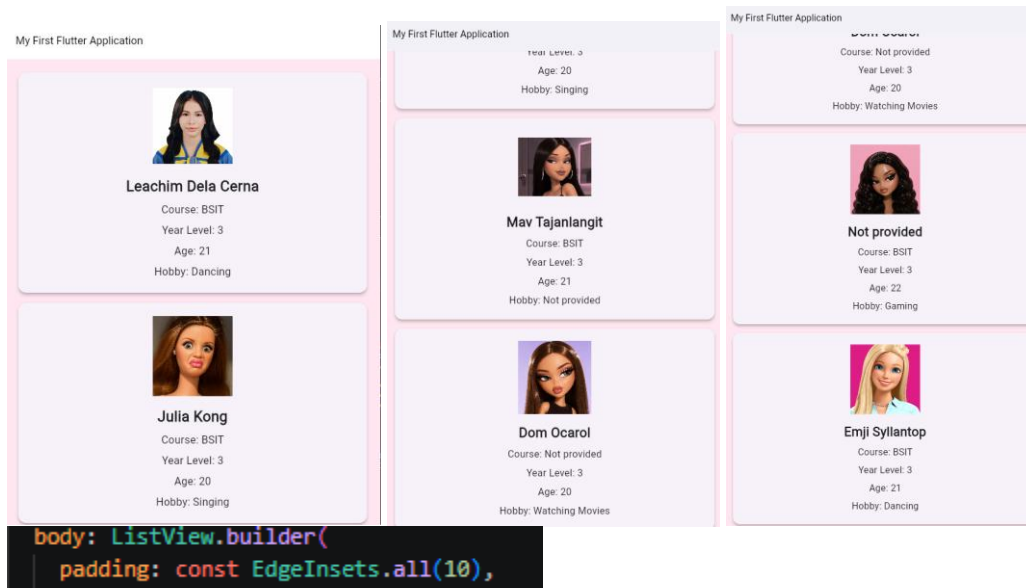
Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
List<Profile> profiles = [
  // STUDENT 1
  Profile(
    image: 'assets/DelaCerna.jpg',
    name: 'Leachim Dela Cerna',
    course: 'BSIT',
    yearLevel: 3,
    age: 21,
    hobby: 'Dancing',
    studentId: '2026001',
    email: 'leachim.delecerna@dbtc-cebu.edu.ph',
    favoriteSubject: 'Programming',
  ),

  // STUDENT 2
  Profile(
    image: 'assets/barbie.jpg',
    name: 'Julia Kong',
    course: 'BSIT',
    yearLevel: 3,
    age: 20,
    hobby: 'Singing',
    studentId: '2026002',
    email: 'julia.kong@dbtc-cebu.edu.ph',
    favoriteSubject: 'Web Development',
  ),

  // STUDENT 3
  Profile(
    image: 'assets/barbiel.jpg',
    name: 'Mav Tajanlangit',
    course: 'BSIT',
    yearLevel: 3,
    age: 21,
    hobby: null,
    studentId: '2026003',
    email: 'mav.tajanlangit@dbtc-cebu.edu.ph',
    favoriteSubject: 'Database',
  ),

  // STUDENT 4
  Profile(
    image: 'assets/barbie3.jpg',
    name: 'Dom Ocarol',
    course: null,
    yearLevel: 3,
    age: 20,
    hobby: 'Watching Movies',
    studentId: '2026004',
    email: 'dom.ocarol@dbtc-cebu.edu.ph',
    favoriteSubject: 'Information Management',
  ),

  // STUDENT 5
  Profile(
    image: 'assets/barbie4.jpg',
    name: 'Krylle Racaza',
    course: 'BSIT',
    yearLevel: 3,
    age: 22,
    hobby: 'Gaming',
    studentId: '2026005',
    email: 'krylle.racaza@dbtc-cebu.edu.ph',
    favoriteSubject: 'Networking',
  ),

  // STUDENT 6
  Profile(
    image: 'assets/emji.jpg',
    name: 'Emji Syllanto',
    course: 'BSIT',
    yearLevel: 3,
    age: 21,
    hobby: 'Dancing',
    studentId: '2026006',
    email: 'emji.syllanto@dbtc-cebu.edu.ph',
    favoriteSubject: 'Programming',
  ),
];
```

My First Flutter Application



Leachim Dela Cerna
 Student ID: 2026001
 Email: leachim.delecerna@dbtc-cebu.edu.ph
 Course: BSIT
 Year Level: 3
 Age: 21
 Hobby: Dancing
 Favorite Subject: Programming



Julia Kong
 Student ID: 2026002
 Email: julia.kong@dbtc-cebu.edu.ph
 Course: BSIT
 Year Level: 3
 Age: 20
 Hobby: Singing
 Favorite Subject: Web Development

My First Flutter Application



Mav Tajanlangit
 Student ID: 2026003
 Email: mav.tajanlangit@dbtc-cebu.edu.ph
 Course: BSIT
 Year Level: 3
 Age: 21
 Hobby: Not provided
 Favorite Subject: Database



Dom Ocarol
 Student ID: 2026004
 Email: dom.ocarol@dbtc-cebu.edu.ph
 Course: Not provided
 Year Level: 3
 Age: 20
 Hobby: Watching Movies
 Favorite Subject: Information Management

My First Flutter Application



Krylle Racaza
 Student ID: 2026005
 Email: krylle.racaza@dbtc-cebu.edu.ph
 Course: BSIT
 Year Level: 3
 Age: 22
 Hobby: Gaming
 Favorite Subject: Networking



Emji Syllanto
 Student ID: 2026006
 Email: emji.syllanto@dbtc-cebu.edu.ph
 Course: BSIT
 Year Level: 3
 Age: 21
 Hobby: Dancing
 Favorite Subject: Programming

Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

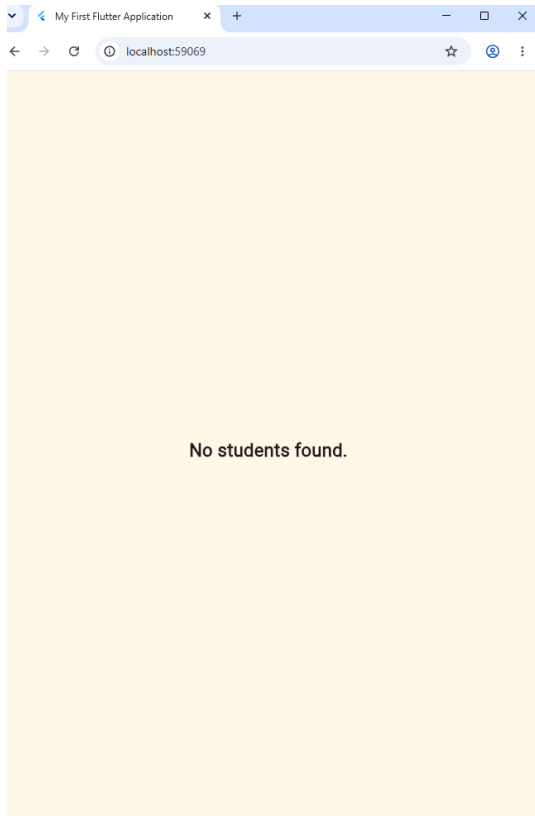
- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list
- Empty-state UI
- Application still functioning correctly



🚩 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a `List`.

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

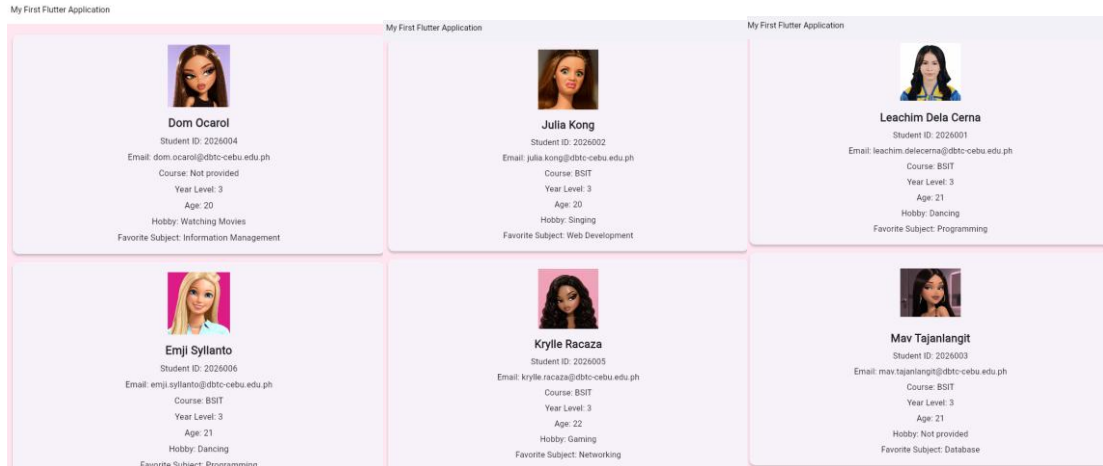
After sorting:

- Aaron
- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
Application finished.
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git branch
  feature/data-ui
* feature/student-list
  feature/widgets
  main
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git status
On branch feature/student-list
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   lib/main.dart
        modified:   pubspec.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        assets/emji.jpg

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git add .
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git commit -m "feat: submit student-list"
[feature/student-list 4473347] feat: submit student-list
 3 files changed, 155 insertions(+), 18 deletions(-)
 create mode 100644 my_first_flutter_app/assets/emji.jpg
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> git push origin feature/student-list
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 16.52 KiB | 16.52 MiB/s, done.
Total 8 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote:   https://github.com/chm0-boop/IT314AB_DelaCerna/pull/new/feature/student-list
remote:
To https://github.com/chm0-boop/IT314AB_DelaCerna.git
 * [new branch]      feature/student-list -> feature/student-list
PS C:\Users\Student\Desktop\IT314AB_DelaCerna\my_first_flutter_app> 
```